

RIS: Main beam toward RX + sidelobe-only environment scan

This note reformulates the idea of **keeping the primary (focal) response aligned to the receiver** while using **dithering** to probe the environment—conceptually scanning **sidelobes** or ambient radiation—without destroying the main link.

1 What you are trying to do

You want a **two-part** surface modulation on the RIS:

Main term	Keeps the RIS-assisted contribution strong and stable at the RX (the “beam” or focus pointed at the RX).
Dither / scan term	Changes how energy is distributed elsewhere (sidelobes / off-focus regions) to sense scatterers, blockers, or changes—ideally with minimal impact on the RX.

Important: on a finite aperture you cannot literally move “only” sidelobes with **zero** effect on the main coupling unless you **constrain** how you dither. The rigorous formulation is: **keep the perturbation small and/or orthogonal to the RX coupling**.

2 Discrete RIS model (schematic)

Model the RIS by N unit cells. Let the **complex spatial modulation** (per cell) be stacked in a vector:

$$\mathbf{\Gamma} = [\Gamma_1, \Gamma_2, \dots, \Gamma_N]^\top \in \mathbb{C}^N. \quad (1)$$

The **scalar** (or dominant) complex gain seen at the RX through the RIS path can be written schematically as an **inner product** with a **coupling vector** from the elements to the RX:

$$h_{\text{RIS}} \propto \mathbf{g}_{\text{rx}}^H \mathbf{\Gamma}. \quad (2)$$

- $\mathbf{g}_{\text{rx}} \in \mathbb{C}^N$ encodes how each element couples to the RX (phases, amplitudes, path aggregation).
- Your **baseline** “point the main toward RX” pattern is some $\mathbf{\Gamma}_0$ chosen so that $\mathbf{g}_{\text{rx}}^H \mathbf{\Gamma}_0$ is large (after any normalization you use).

This is the right abstraction for both **far-field steering** (where $\mathbf{g}_{\text{rx}}$ looks like a plane-wave array response) and **near-field focusing** (where $\mathbf{g}_{\text{rx}}$ includes **range and curvature**, not just angle).

3 Splitting main and dither

3.1 Additive complex perturbation

$$\mathbf{\Gamma}_k = \mathbf{\Gamma}_0 + \Delta\mathbf{\Gamma}_k. \quad (3)$$

- $\mathbf{\Gamma}_0$: fixed (or slowly tracked) **main** pattern toward the RX.
- $\Delta\mathbf{\Gamma}_k$: k -th **dither** pattern for environment scanning.

You usually **renormalize** $\mathbf{\Gamma}_k$ to satisfy hardware constraints (e.g. per-element magnitude limits).

3.2 Phase-only dither (common for passive RIS)

$$\Gamma_{k,n} = \Gamma_{0,n} e^{j\delta_{k,n}}, \quad n = 1, \dots, N. \quad (4)$$

- $\delta_k = [\delta_{k,1}, \dots, \delta_{k,N}]^T$: small or structured **phase ripple** on top of the main profile.

4 Two ways to get “main stays, environment is probed”

4.1 Null-space / orthogonal dither (strongest interpretation)

You want $\Delta\Gamma_k$ to **not** move the RX coupling, in first order:

$$\mathbf{g}_{\text{rx}}^H \Delta\Gamma_k \approx 0. \quad (5)$$

So $\Delta\Gamma_k$ lies approximately in the **orthogonal complement** of $\mathbf{g}_{\text{rx}}$.

Construction (conceptual):

1. Form $\mathbf{g}_{\text{rx}}$ (from geometry + your near-field or far-field model, or from a simulator).
2. Build an orthonormal basis $\{\mathbf{u}_1, \dots, \mathbf{u}_M\}$ orthogonal to $\mathbf{g}_{\text{rx}}$:

$$\mathbf{g}_{\text{rx}}^H \mathbf{u}_m = 0, \quad m = 1, \dots, M. \quad (6)$$

3. Define dither codewords as

$$\Delta\Gamma_k = \sum_{m=1}^M \alpha_{k,m} \mathbf{u}_m, \quad (7)$$

with small coefficients $\alpha_{k,m}$.

4. Full pattern:

$$\Gamma_k = \text{Normalize}(\Gamma_0 + \Delta\Gamma_k). \quad (8)$$

Intuition: the RX inner product stays dominated by Γ_0 ; the dither mostly changes **how energy is distributed away from** the RX coupling—what you loosely call **sidelobes** or non-focal radiation.

4.2 Small high-spatial-frequency ripple (simpler, less controlled)

Add a **weak, rapidly varying** phase δ_k on top of Γ_0 . Often leaves the coherent sum toward RX mostly unchanged if ripple is small and balanced; still perturbs off-boresight structure. Easier to implement than null-space projection but typically causes **some** main-link variation.

5 What the near-field case changes

- **Far-field:** $\mathbf{g}_{\text{rx}}$ is often well approximated by a **plane-wave** array steering vector.
- **Near-field (Fresnel region):** $\mathbf{g}_{\text{rx}}$ must reflect **distance and wavefront curvature** across elements—not only angle.

So **main toward RX** should be **near-field-aware** Γ_0 , and **null-space dither** should be orthogonal to this **near-field coupling vector**, not only to a far-field steering vector.

6 Practical recipe

1. Define $\mathbf{g}_{\text{rx}}$ for your layout.
2. Set $\mathbf{\Gamma}_0$ to maximize $|\mathbf{g}_{\text{rx}}^H \mathbf{\Gamma}_0|$ (often $\mathbf{\Gamma}_0 \propto \mathbf{g}_{\text{rx}}^*$ up to normalization).
3. Build basis $\{\mathbf{u}_m\}$ with $\mathbf{g}_{\text{rx}}^H \mathbf{u}_m = 0$.
4. Form $\mathbf{\Gamma}_k = \text{Normalize}(\mathbf{\Gamma}_0 + \sum_m \alpha_{k,m} \mathbf{u}_m)$ with small $\alpha_{k,m}$.
5. Verify $|\mathbf{g}_{\text{rx}}^H \mathbf{\Gamma}_k|$ is stable across k .

7 Caveats

1. **Exact** “no change on main” only holds in a **linear** sense; large phase-only dither still couples nonlinearly through $\exp(j\delta)$.
2. **Normalization** can couple dither and main when projecting to feasible $\mathbf{\Gamma}_k$.
3. In Sionna-style simulators you implement $\mathbf{\Gamma}_k$ as your own profiles on the RIS grid; this math is **your codebook design**.

8 One-line summary

Keep a near-field-correct main pattern $\mathbf{\Gamma}_0$ aligned to the RX; add dither $\Delta\mathbf{\Gamma}_k$ that is approximately orthogonal to $\mathbf{g}_{\text{rx}}$ so the primary link is preserved while the rest of the field is modulated for environment sensing.